

# Germanic lexeme report assembly pilot 01

## Representative model-entry assembly test

This pilot assembles a small representative subset of the current Germanic model entries for publication-format testing. Each entry combines the compact transducer trace with the current lexeme-report prose, while leaving the source `.model.md` files unchanged.

The included entries are listed in `pilot_manifest.tsv` and appear here in stable manifest order. Compact trace blocks come from `oe_derivation_class_trace_report.compact.md`; philological and literature discussion comes from the current model-entry prose. Implementation reports, reviewer checklists, source ledgers, packets, and research notes are intentionally excluded from the assembled body.

### bake — OE *bacan*

| Item                      | Value                          |
|---------------------------|--------------------------------|
| lexical item              | bake                           |
| citation reconstruction   | <i>*bákanq</i>                 |
| selected transducer input | <i>*bákanq</i>                 |
| Old English target        | <i>bacan</i>                   |
| derivation class          | regular                        |
| transducer output         | <i>*bákanq</i> -> <i>bacan</i> |

### Derivation trace

Proto input: *\*bákanq*

| Earlier Germanic changes | Old English changes                      |
|--------------------------|------------------------------------------|
| West Germanic            | Old English                              |
| [no change]              | Anglo Frisian Brightening <i>*bækanq</i> |
| Northwest Germanic       | OE A Restoration <i>*bakanq</i>          |
| [no change]              | OE Heavy Syllable Nasal <i>*bakan</i>    |
|                          | Apocope                                  |
|                          | OE Secondary Nasalization <i>*bakqn</i>  |
|                          | OE Weak Tail Reduction <i>*bakan</i>     |

Outcome: *bacan*

### Reconstruction and comparative evidence

Orel reconstructs the verb as *\*bakanan* and cites Old English *bacan* beside Old High German *backan*, *bahhan* (Orel 2003). Campbell gives *bacan* as one of the standard examples of Old English A-restoration before a single consonant, and Ringe and Taylor state the same development from *\*bakan* to Old English *bacan* (Campbell 1959, 61; Ringe and Taylor 2014).

## Old English evidence

Bosworth-Toller and Clark Hall both record *bacan* as the ordinary Old English verb ‘to bake’ (Bosworth and Toller 1898, 72; Clark Hall 1960). The target in this entry is therefore the attested infinitive headword itself, not a selected oblique or finite paradigm cell.

## Development to Old English

From *\*bákanq*, Anglo-Frisian brightening first gives *\*bækanq*. A-restoration then returns the stem vowel to *a* before single *k* plus the back-vocalic infinitive suffix, and later apocope and weak-tail reduction yield *bacan* (Campbell 1959, 61; Ringe and Taylor 2014). The development is therefore straightforward: *\*bákanq* > *bacan*.

## bow — OE *bēag*

| Item                      | Value                       |
|---------------------------|-----------------------------|
| lexical item              | bow                         |
| citation reconstruction   | <i>*béuganq</i>             |
| selected transducer input | <i>*báug</i>                |
| Old English target        | <i>bēag</i>                 |
| derivation class          | late_analogy                |
| transducer output         | <i>*báug</i> -> <i>bēag</i> |

## Derivation trace

Proto input: *\*báug*

| Earlier Germanic changes | Old English changes           |
|--------------------------|-------------------------------|
| West Germanic            | Old English                   |
| [no change]              | OE Au Fronting                |
| Northwest Germanic       | OE Diphthong Leveling         |
| [no change]              | <i>*báeug</i><br><i>*bēag</i> |

Outcome: *bēag*

## Reconstruction and comparative evidence

The inherited verb belongs to the class-II strong-verb family *\*béuganq* (Ringe and Taylor 2014, 55). Within that paradigm, however, the infinitive and the singular preterite continue different ablaut grades. The selected input *\*báug* is the singular preterite cell, whereas the citation form *\*béuganq* is the infinitive.

Campbell’s account of Old English class-II strong verbs treats the singular preterite *au* > *ēa* development as regular in this environment (Campbell 1959, 53). That is the phonological path relevant for *bēag*, whereas the analogical *ū* of the present stem belongs to the separate history behind *būgan* (Ringe and Taylor 2014, 55).

## Old English evidence

Bosworth-Toller and Clark Hall both record *bēag* as a preterite form of *būgan* (Bosworth and Toller 1898, 122; Clark Hall 1960, 45). The form discussed here is therefore an attested Old English verbal form, not a reconstructed substitute for the infinitive.

The ordinary dictionary headword remains *būgan*, but the relevant comparison form for this entry is the singular preterite *bēag*. That is the paradigm cell in which the inherited *\*au* grade is preserved most directly.

## Development to Old English

From *\*báug*, Anglo-Frisian fronting and the later leveling of the diphthong produce *bēag* (Campbell 1959, 53). No special analogical repair is needed for that cell. The form is the regular Old English outcome of the singular-preterite grade.

The analogical element in the wider lexeme belongs instead to the present stem seen in *būgan*. The selected input differs from the citation form because the regular inherited pathway survives more transparently in the preterite than in the infinitive.

## Paradigm comparison

The comparison below is manual. It distinguishes the regular singular preterite from the more familiar infinitival citation form.

| PGmc cell / interpretation | Candidate input                 | Expected or documented OE outcome                  | OE comparison form  | Result                                                              |
|----------------------------|---------------------------------|----------------------------------------------------|---------------------|---------------------------------------------------------------------|
| citation infinitive        | <i>*béuganaŋ</i>                | inherited present-stem history behind <i>būgan</i> | <i>būgan</i>        | establishes the lexeme, but not the selected target                 |
| singular preterite         | <i>*báug</i>                    | compact-trace output: <i>bēag</i>                  | <i>bēag</i>         | exact match between input, output, and attested cell                |
| past participial branch    | participial <i>*bugan-</i> type | later participial outcomes                         | bogen-type evidence | relevant to the paradigm, but not the clearest match for this entry |

The singular preterite is the relevant comparison form. It gives a direct lautgesetzlich path to attested *bēag*, while the citation form *būgan* belongs to a paradigm whose present stem has already undergone later leveling.

## craft — OE cræft

| Item                      | Value                           |
|---------------------------|---------------------------------|
| lexical item              | craft                           |
| citation reconstruction   | <i>*kráftiz</i>                 |
| selected transducer input | <i>*kráftaz</i>                 |
| Old English target        | <i>cræft</i>                    |
| derivation class          | early_analogy                   |
| transducer output         | <i>*kráftaz</i> -> <i>cræft</i> |

## Derivation trace

Proto input: *\*kráftaz*

| Earlier Germanic changes |                | Old English changes       |               |
|--------------------------|----------------|---------------------------|---------------|
| West Germanic            |                | Old English               |               |
| [no change]              |                | PWGmc Final Bare A Loss   | <i>*kráft</i> |
| Northwest Germanic       |                | Anglo Frisian Brightening | <i>*kræft</i> |
| PGmc Final Z Deletion    | <i>*kráfta</i> |                           |               |

Outcome: *cræft*

## Reconstruction and comparative evidence

Comparative sources disagree about the older stem class. Kroonen gives a u-stem *\*kraftu-*, while Orel prints *\*kraftiz* *\*kraftuz* (Kroonen 2013, 340; Orel 2003, 259). The comparative label *\*kráftiz* remains in view as a lexeme-level shorthand, while *\*kráftaz* is the pre-Old-English form used for the Old English derivation.

## Old English evidence

The Old English noun itself is secure. Clark Hall and Bosworth-Toller both give *cræft* as the head-word (Clark Hall 1960, 19; Bosworth and Toller 1898, 145).

## Development to Old English

The comparison is between possible pre-Old-English inputs. The i-stem comparator *\*kráftiz* gives *creft*, while the u-stem comparator *\*kráftuz* gives *craft*. The a-stem-shaped input *\*kráftaz* yields *cræft* and is therefore the form used for the Old English derivation. This does not require treating the comparative dictionaries as identical; it shows the narrower point that the Old English derivation needs a pre-Old-English form without the i-umlaut trigger of *\*-iz* and without the back-vowel outcome associated with the u-stem comparator.

## Form comparison

| Candidate input | OE output    | Result                             |
|-----------------|--------------|------------------------------------|
| *kráftiz        | <i>creft</i> | non-match; i-stem comparator       |
| *kráftuz        | <i>craft</i> | non-match; u-stem comparator       |
| *kráftaz        | <i>cræft</i> | exact match; selected pre-OE input |

## fowl — OE fugol

| Item                      | Value                  |
|---------------------------|------------------------|
| lexical item              | fowl                   |
| citation reconstruction   | *fúglaz                |
| selected transducer input | *fúglaz                |
| Old English target        | fugol                  |
| derivation class          | unexplained_unmodelled |
| transducer output         | *fúglaz -> fogol       |

## Derivation trace

Proto input: *\*fúglaz*

| Earlier Germanic changes |         | Old English changes     |        |
|--------------------------|---------|-------------------------|--------|
| West Germanic            |         | Old English             |        |
| [no change]              |         | PWGmc Final Bare A Loss | *fógl  |
| Northwest Germanic       |         | OE Epenthetic Vowel     | *fógol |
| NWGmc U Lowering         | *fóglaz |                         |        |
| PGmc Final Z Deletion    | *fóglā  |                         |        |

Transducer outcome: *fogol*

Selected target: *fugol*

## Reconstruction and comparative evidence

The noun is the ordinary Germanic a-stem *\*fúglaz*, continued by forms such as Old Norse *fugl* and Old High German *fogal* (Kroonen 2013, 197; Orel 2003, 155). There is no stem-class or paradigm-cell dispute behind this entry. The comparative headword and the selected input are the same.

The difficulty lies instead in the stressed root vowel. Under the regular West Germanic and Old English development, that *u* should lower before the following non-high vowel, yielding an *o*-vocalism (Ringe and Taylor 2014, 42–43; Campbell 1959, 43).

## Old English evidence

Old English dictionaries record the noun as *fugol*, with variant spelling *fugel* (Bosworth and Toller 1898, 282; Clark Hall 1960, 138). The target is therefore an attested ordinary Old English noun, not a reconstructed or selectively chosen paradigm form.

The attested word already contains the crucial problem. Its medial *-o-* is the ordinary parasite vowel of Old English cluster phonology, but the root *fū-* retains *u* where the regular history predicts *fo-* (Campbell 1959, 150).

## Development to Old English

From *\*fúglaz*, the regular cascade yields *fogol*: the root vowel lowers before the following non-high vowel (Ringe and Taylor 2014, 42–43; Campbell 1959, 43), final *z* is lost, and the cluster is resolved by the usual medial vowel (Ringe and Taylor 2014, 345; Campbell 1959, 150). That is the expected inherited outcome.

The attested Old English noun is *fugol*, not *fogol*. Luick and later handbooks treat this preservation of *u* as a small inherited residue, not as a categorical sound law (Luick 1914–1940, 148; Ringe and Taylor 2014, 47). The item therefore remains a genuine lexical exception rather than the output of a recoverable regular mechanism.

## Expected and attested forms

The comparison below is manual. It distinguishes the regular prediction from the attested Old English noun.

| Form         | Status                                      | Relevance to this entry                              |
|--------------|---------------------------------------------|------------------------------------------------------|
| <i>fogol</i> | computed regular output from <i>*fúglaz</i> | establishes the expected inherited outcome           |
| <i>fugol</i> | attested Old English form                   | selected target; preserves unexplained root <i>u</i> |
| <i>fugel</i> | attested variant spelling                   | secondary spelling variant of the attested noun      |

The unresolved point lies only in the root vowel. The medial *-o-* is regular, but no accepted lautgesetzlich pathway has been found from *\*fúglaz* to attested *fugol*.

## thistle — OE pistles

| Item                      | Value            |
|---------------------------|------------------|
| lexical item              | thistle          |
| citation reconstruction   | <i>*θéstilaz</i> |
| selected transducer input | <i>*θístilas</i> |
| Old English target        | <i>pistles</i>   |
| derivation class          | late_analogy     |

| Item              | Value                              |
|-------------------|------------------------------------|
| transducer output | <i>*θístilas</i> -> <i>pistles</i> |

## Derivation trace

Proto input: *\*θístilas*

| Earlier Germanic changes |                    | Old English changes       |                  |
|--------------------------|--------------------|---------------------------|------------------|
| West Germanic            |                    | Old English               |                  |
| [no change]              |                    | Anglo Frisian Brightening | <i>*θístilæs</i> |
|                          | Northwest Germanic | OE L Adjacent Syncope     | <i>*θístlæs</i>  |
| [no change]              |                    | OE Unstressed AE Merger   | <i>*θístles</i>  |

Outcome: *pistles*

## Reconstruction and comparative evidence

The comparative tradition is divided. Orel prints *\*þe(x)stilaz*, while Kluge-Seebold gives *\*þistila-* (Orel 2003, 458; Kluge 2011). The comparative label *\*θéstilaz* therefore remains in view as the lexeme-level headword, while the selected input *\*θístilas* is a specific genitive singular cell.

## Old English evidence

The ordinary simplex headword tradition is broken *pistel* / *ðistel*. Clark Hall gives *ðistel* as the noun headword (Clark Hall 1960, 326). The selected target here is the genitive singular *pistles*, which preserves the same stem in an oblique form where the cluster is medial.

## Development to Old English

Campbell's discussion of cluster nouns shows the contrast clearly. Simplex forms often develop a parasite vowel in word-final obstruent + sonorant clusters, while comparable medial clusters remain unbroken; his examples include *hrefn*, *tacn*, *wépn*, and *botm* beside forms with parasitic vowels elsewhere in the same lexical class (Campbell 1959, 151). The selected genitive singular *\*θístilas* therefore supplies the conservative comparison form: the cluster is medial and the regular development yields *pistles*, while the simplex nominative belongs to the broken headword tradition *pistel*.

## Paradigm comparison

The comparison below is manual. It shows the contrast between the citation form and the selected genitive singular cell.

| PGmc cell / interpretation   | Candidate input  | OE output or comparison         | OE comparison form | Result                                                       |
|------------------------------|------------------|---------------------------------|--------------------|--------------------------------------------------------------|
| citation nominative singular | <i>*θéstilaz</i> | computed output: <i>pistl</i>   | <i>pistel</i>      | useful citation-form background, but not the selected target |
| selected genitive singular   | <i>*θístilas</i> | computed output: <i>pistles</i> | <i>pistles</i>     | exact match for the chosen conservative cell                 |

## weapon — OE *wǣpn*

| Item                      | Value                         |
|---------------------------|-------------------------------|
| lexical item              | weapon                        |
| citation reconstruction   | * <i>wépnq</i>                |
| selected transducer input | * <i>wépnq</i>                |
| Old English target        | <i>wǣpn</i>                   |
| derivation class          | regular                       |
| transducer output         | * <i>wépnq</i> -> <i>wǣpn</i> |

### Derivation trace

Proto input: \**wépnq*

| Earlier Germanic changes |                | Old English changes             |               |
|--------------------------|----------------|---------------------------------|---------------|
| West Germanic            |                | Old English                     |               |
| [no change]              |                | OE Heavy Syllable Nasal Apocope | * <i>wǣpn</i> |
| Northwest Germanic       |                |                                 |               |
| NWGmc Long E Lowering    | * <i>wǣpnq</i> |                                 |               |

Outcome: *wǣpn*

### Reconstruction and comparative evidence

Kroonen reconstructs a double-stem noun \**wēbna-* ~ \**wēpna-* and cites OE *wǣpn* among its reflexes (Kroonen 2013, 617). The selected input \**wépnq* represents the unbroken citation-form noun rather than the later broken simplex.

### Old English evidence

Campbell's cluster-noun discussion preserves unbroken *wépn* beside broken *wépen*-type forms (Campbell 1959, 150; 1959, 226–27). Bright contrasts broken nominative *wǣpen/wapen* with unbroken oblique *wǣpnes*, while Clark Hall lemmatizes the noun under *wapen* and also preserves unbroken forms in compounds and related spellings (Bright 1971, 29; Clark Hall 1960, 355).

### Development to Old English

Northwest Germanic lowering gives *wǣpn*, and loss of the final nasal vowel leaves the unbroken cluster word-finally. The selected target is the attested unbroken form *wǣpn*.

### Form note

The ordinary late West Saxon simplex headword is *wǣpen*, but Campbell's noun-class discussion also preserves unbroken *wépn* beside broken *wépen*-type forms (Campbell 1959, 150; 1959, 226–27). *wǣpnes* remains the regular unbroken oblique comparator (Bright 1971, 29; Clark Hall 1960, 355).

## will — OE *willa*

| Item                      | Value          |
|---------------------------|----------------|
| lexical item              | will           |
| citation reconstruction   | * <i>wéljô</i> |
| selected transducer input | * <i>wéljô</i> |
| Old English target        | <i>willa</i>   |

| Item              | Value           |
|-------------------|-----------------|
| derivation class  | regular         |
| transducer output | *wéljô -> willa |

### Derivation trace

Proto input: \*wéljô

| Earlier Germanic changes |         | Old English changes                 |         |
|--------------------------|---------|-------------------------------------|---------|
| West Germanic            |         | Old English                         |         |
| PWGmc J Gemination       | *wélljô | OE I Umlaut                         | *willjô |
| Northwest Germanic       |         | OE Unstressed Long Vowel Shortening | *willja |
| [no change]              |         | OE J Loss After Heavy               | *willa  |

Outcome: willa

### Reconstruction and comparative evidence

Kroonen separates noun \*weljan- 2 ‘will, wish’ from verb \*weljan- 1 ‘to want’, while Orel and Kluge represent the noun as \*weljôn/\*weljOn (Kroonen 2013; Orel 2003; Kluge 2011). The selected derivational form \*wéljô is the noun-side input used for this row.

### Old English evidence

Clark Hall lemmatizes noun *willa m.* separately from verb *willan* (Clark Hall 1960, 368). The selected target is the noun citation form, not the related verb.

### Development to Old English

From \*wéljô, j-gemination yields a heavy stem, i-umlaut gives *will-*, and later shortening plus j-loss produce *willa*. The noun is therefore a regular weak masculine outcome.

### Lexical note

The target here is the noun *willa* ‘will, wish’. Related verb *willan* belongs to a separate lexeme and should not be substituted for the noun row (Kroonen 2013; Clark Hall 1960, 368).

## youth — OE *geogup*

| Item                      | Value             |
|---------------------------|-------------------|
| lexical item              | youth             |
| citation reconstruction   | *júgunθiz         |
| selected transducer input | *júgunθ           |
| Old English target        | geogup            |
| derivation class          | early_analogy     |
| transducer output         | *júgunθ -> geogup |

### Derivation trace

Proto input: \*júgunθ



| Earlier Germanic changes        |          | Old English changes                 |         |
|---------------------------------|----------|-------------------------------------|---------|
| West Germanic                   |          | Old English                         |         |
| [no change]                     |          | OE Diphthong Leveling               | *jéogūθ |
|                                 |          | OE Unstressed Long Vowel Shortening | *jéoguθ |
| Northwest Germanic              |          |                                     |         |
| OE Ws Palatal Glide             | *jéugunθ |                                     |         |
| NWGmc Nasal Spirant Lengthening | *jéugūnθ |                                     |         |
| NWGmc Nasal Spirant Loss        | *jéugūθ  |                                     |         |

Outcome: *geogub*

### Reconstruction and comparative evidence

The wider etymological tradition reconstructs an earlier form of the word as *\*ju(w)unþi-* (Kroonen 2013; Fulk 2018). The selected comparative label *\*júgunθiz* already stands at a later Germanic stage with *g*, and the chosen input *\*júgunθ* is later again: it represents the form after final *-i* has been lost.

That staging matters because Ringe and Taylor explicitly give the sequence *\*jugunþi* > *\*jugub* > *OE geogub* ~ *iugub* (Ringe and Taylor 2014). Campbell makes the same point with the parallel *dugub* < *\*dugunþ-*, adding *geogub* to the same history (Campbell 1959). The selected input therefore differs from the broader comparative headword because the Old English development must begin after early loss of final *-i*.

### Old English evidence

The Old English noun is attested with varying spellings. Campbell records forms of the *iugub* / *gioguð* / *geoguð* type, and Ringe and Taylor likewise cite *geogub* ~ *iugub* (Campbell 1959; Ringe and Taylor 2014). The form is normalized here as *geogub*: the initial palatal is written with *g*, and the attested spelling variation is treated as orthographic rather than lexical.

Nothing in the source stack suggests that a different paradigm cell should be chosen. The relevant Old English comparison form is the noun *geogub* itself.

### Development to Old English

The decisive early step is the loss of final *-i* before the Old English umlaut stage. If that high vowel remained, the word would develop an over-umlauted *y*-type vowel instead of the attested form (Ringe and Taylor 2014).

From the selected input *\*júgunθ*, the later development is regular: palatal fronting yields *\*jéugunθ*; nasal-spirant lengthening and loss give *\*jéogūθ*; unstressed long-vowel shortening then produces *\*jéoguθ*, which surfaces as *geogub*. Medial *u* remains preserved because the handbooks treat this environment as one of stem-*u* harmony after stressed *u*, citing forms such as *munuc*, *dugub*, and *iugub* (Campbell 1959; Sievers 1965; Luick 1914–1940, 397).

### Stage comparison

The comparison below is manual. It separates the broader comparative headword from the later stages relevant to the Old English noun.

| Stage / interpretation        | Candidate form     | Old English outcome or comparison | Relevance to this entry                        |
|-------------------------------|--------------------|-----------------------------------|------------------------------------------------|
| earlier etymological headword | <i>*ju(w)unþi-</i> | comparative family background     | older comparative reconstruction of the lexeme |

| Stage / interpretation            | Candidate form | Old English outcome or comparison      | Relevance to this entry                                                     |
|-----------------------------------|----------------|----------------------------------------|-----------------------------------------------------------------------------|
| later g-bearing comparative label | *júgunθiz      | citation reconstruction / lexeme label | preserves the later Germanic stage behind the selected entry                |
| selected OE-facing input          | *júgunθ        | compact-trace output: <i>geogup</i>    | exact match for the chosen Old English form                                 |
| full -i stage retained too long   | *jugunþi       | expected over-umlauted y-type result   | negative control showing why early -i loss must precede the OE umlaut stage |

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